FabCon Atlanta
2026

Module 2 – Outbound Security

Module 2 Instructions – Outbound Protection

Overview

In this module, you will be configuring outbound access protection (OAP) capabilities – Managed Private endpoints (MPE) and Connection Policies.

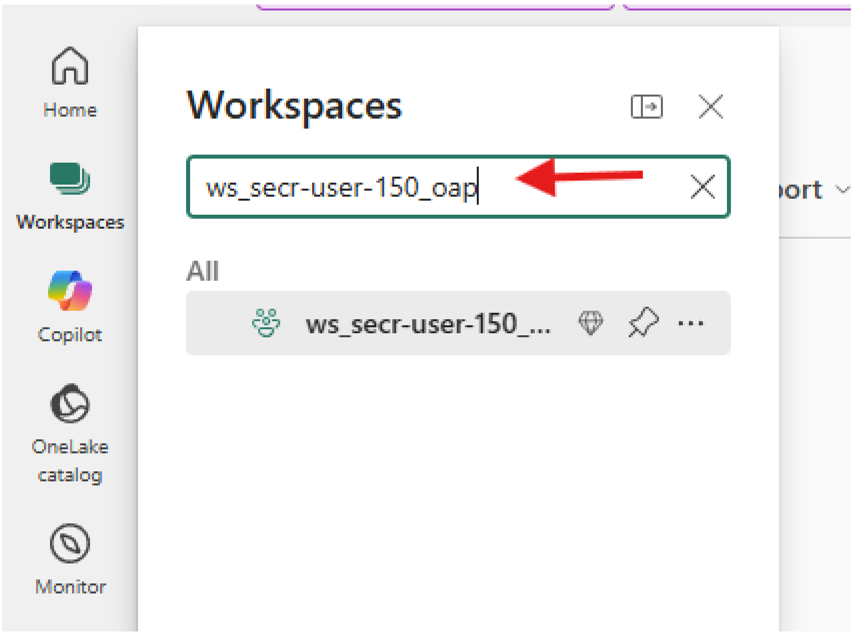
Once configured, you will then test outbound connectivity to write to an external Azure Storage account and another workspace in the same tenant.

For this exercise, you will use pre-configured workspace **ws_*<user>*_oap.**

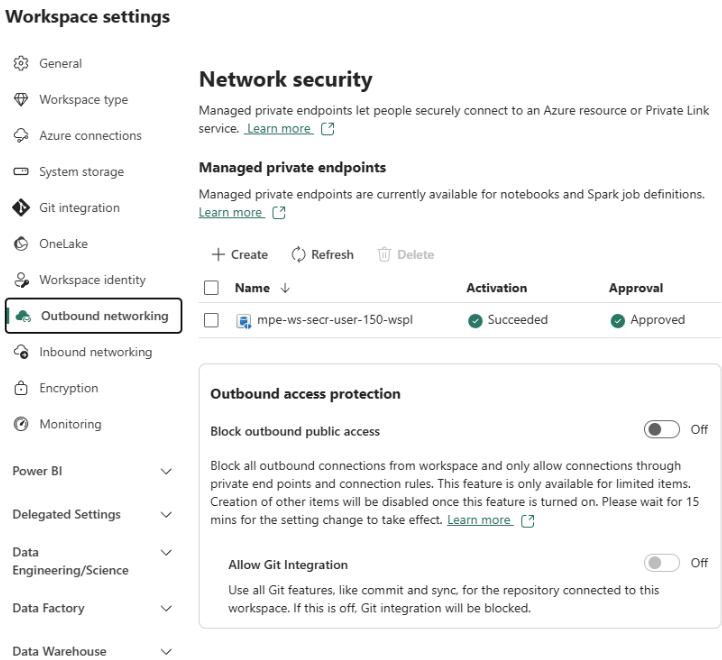
Hands-on: Setup managed private endpoint and connect to ADLS g2 from OAP workspace

Step 1: Enable OAP and configure managed private endpoint (MPE) for ADLS.

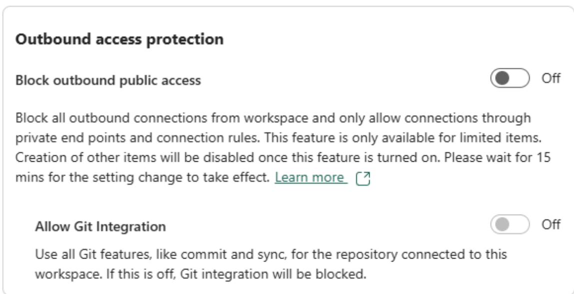
1. Go to Fabric workspace and navigate to pre-provisioned workspace for OAP.



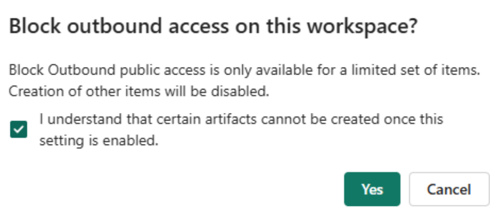
2. Navigate to **Workspace Settings** > **Outbound networking**.



3. Enable outbound access protection for the workspace using the toggle. Once enabled, all outbound connections by supported Fabric workloads will be blocked by default.

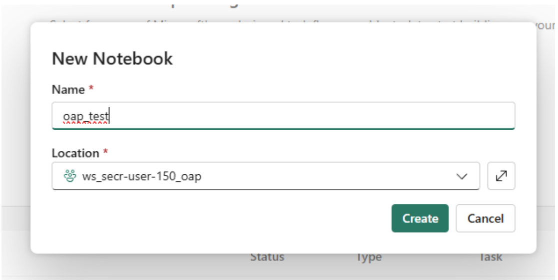


4. Change this toggle to ON; when prompted, mark the checkbox and click on **Yes.**

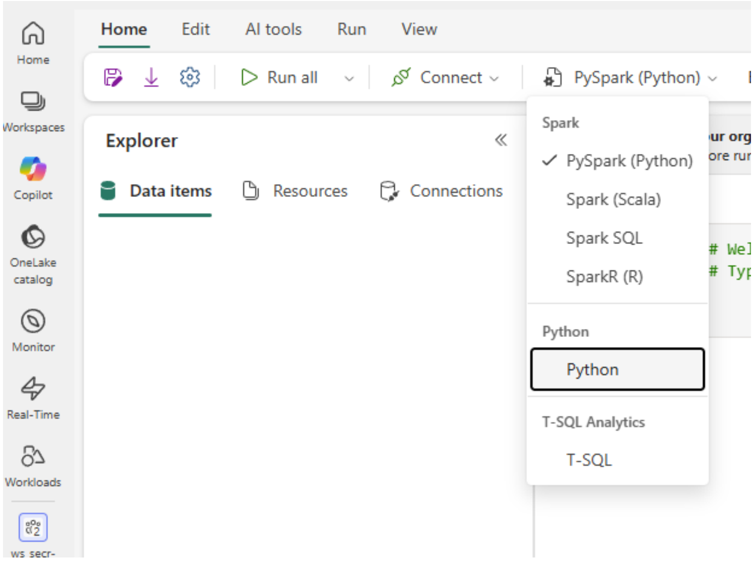


5. Once OAP is enabled for the workspace, you can test writing to an ADLS account using a notebook without a managed private endpoint (MPE).

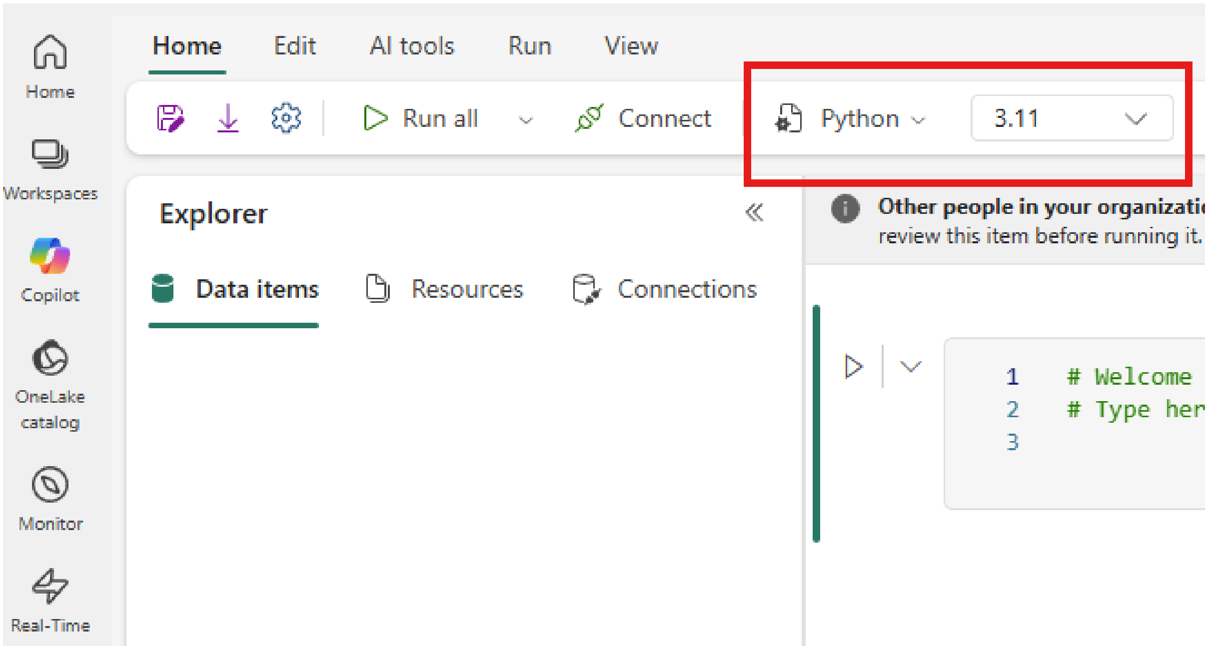
6. Create a New Item of type Notebook called “oap_test”



7. Change the notebook type from PySpark(Python) to Python



Python notebooks are generally faster to start up than PySpark notebooks.



8. Paste the below code snipped (**replace the container with your username**) in a new cell and run. This should fail as outbound is blocked from this workspace.

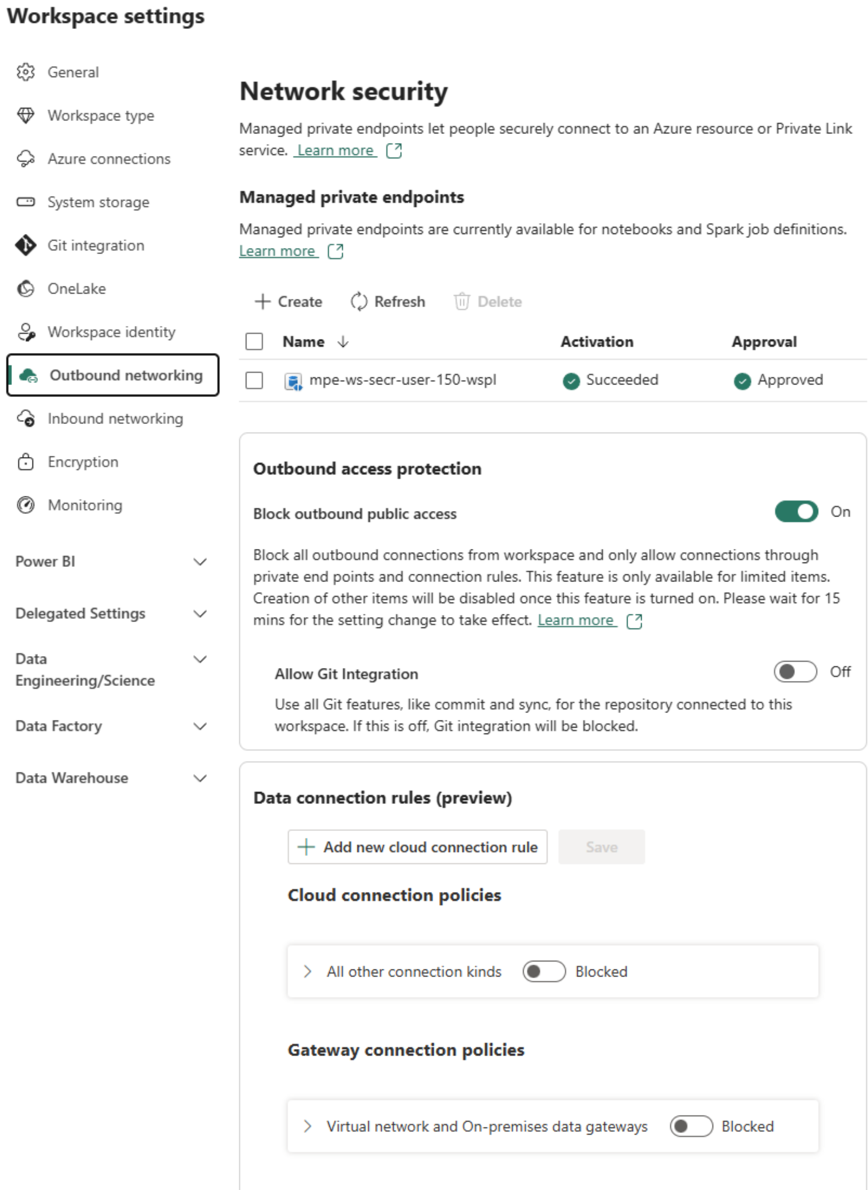
```
import notebookutils

container = "secr-user-150" #replace this with your username
storageaccount = "adlsfabconsec"
directory = "fabric_mpe_test"

test_message = "This is a sample created for testing MPE."
```

```
file_name = (notebookutils.runtime.context['userName']).lower().replace("
","") + ".txt"
adls_path =
f"abfss://{container}@{storageaccount}.dfs.core.windows.net/{directory}/{file_
name}"

#This will write file to the ADLS path.

notebookutils.fs.put(adls_path, test_message, True)
```

9. With outbound connectivity blocked and absence of a managed private endpoint (MPE) to ADLS, the code will fail with the following error.

10. You will now configure an MPE to external ADLS account. Navigate to **Workspace Settings** for the workspace ending with _*oap*.

11. Navigate to **Outbound networking** and then select **+Create** under **Managed private endpoints**.

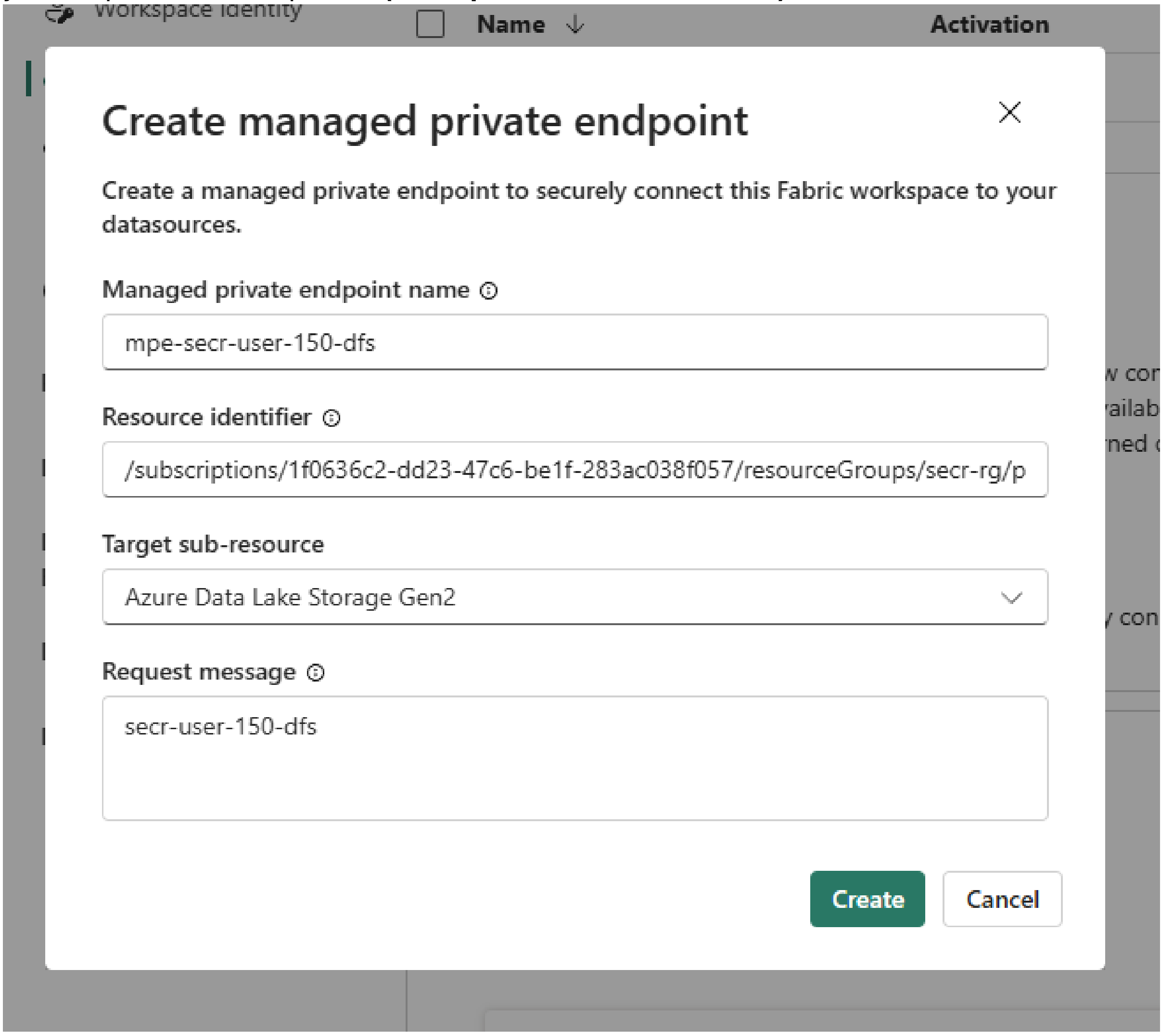


12. Create a new Managed Private Endpoint and fill out the details for the managed private endpoint. You can use the following values for configuring MPE.
    - **Managed private endpoint name: mpe-username-dfs** (example: mpe-secr-user-150-dfs)
    - **Resource identifier:** This is the resource ID from Azure. In this workshop we will be sharing a single ADLS account so you can use the following resource ID value.

      /subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/providers/Microsoft.Storage/storageAccounts/adlsfabconsec

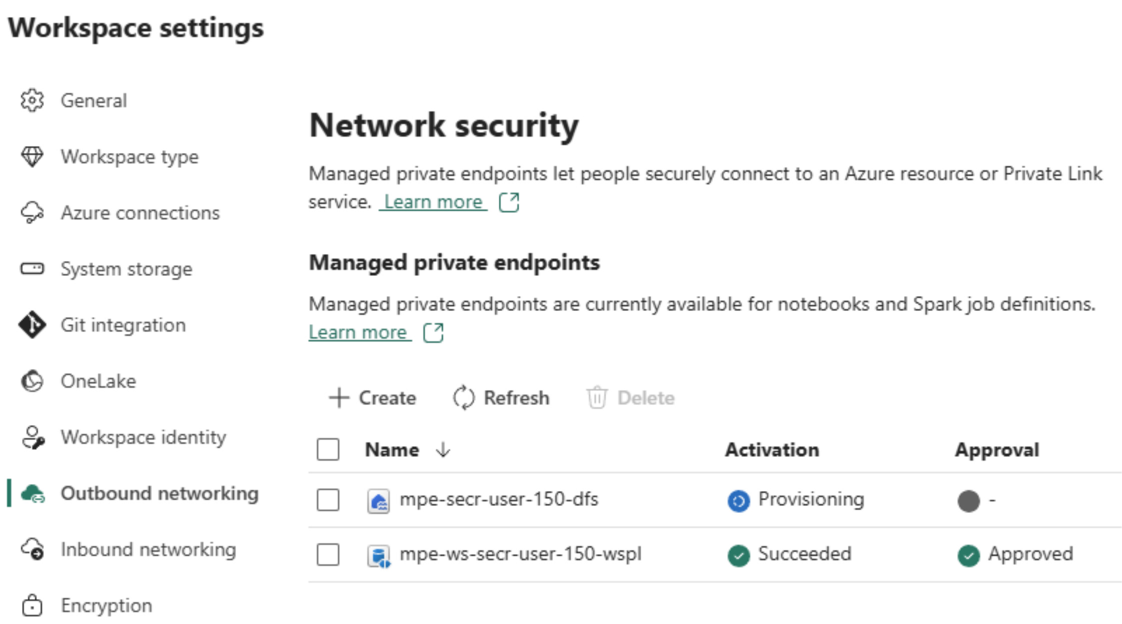
    - **Target sub-resource:** Azure Data Lake Storage Gen2

- **Request message:** This is text field but for this workshop, please put down your username-dfs here. This will make it easier for you to identify your request in Azure portal. **(Example: secr-user-150-dfs)**

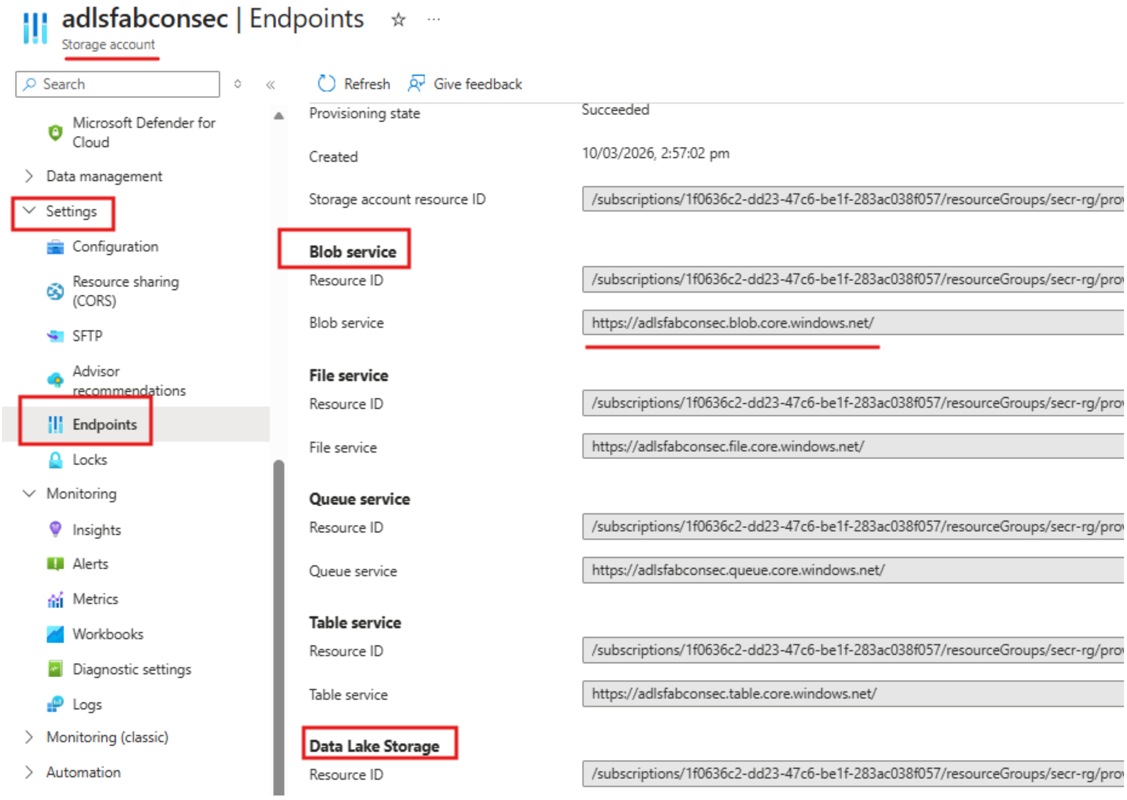


Once details have been entered, click **Create.**

13. **You will see that the MPE is provisioning, wait for 5 minutes, in the meantime, proceed to next step.**



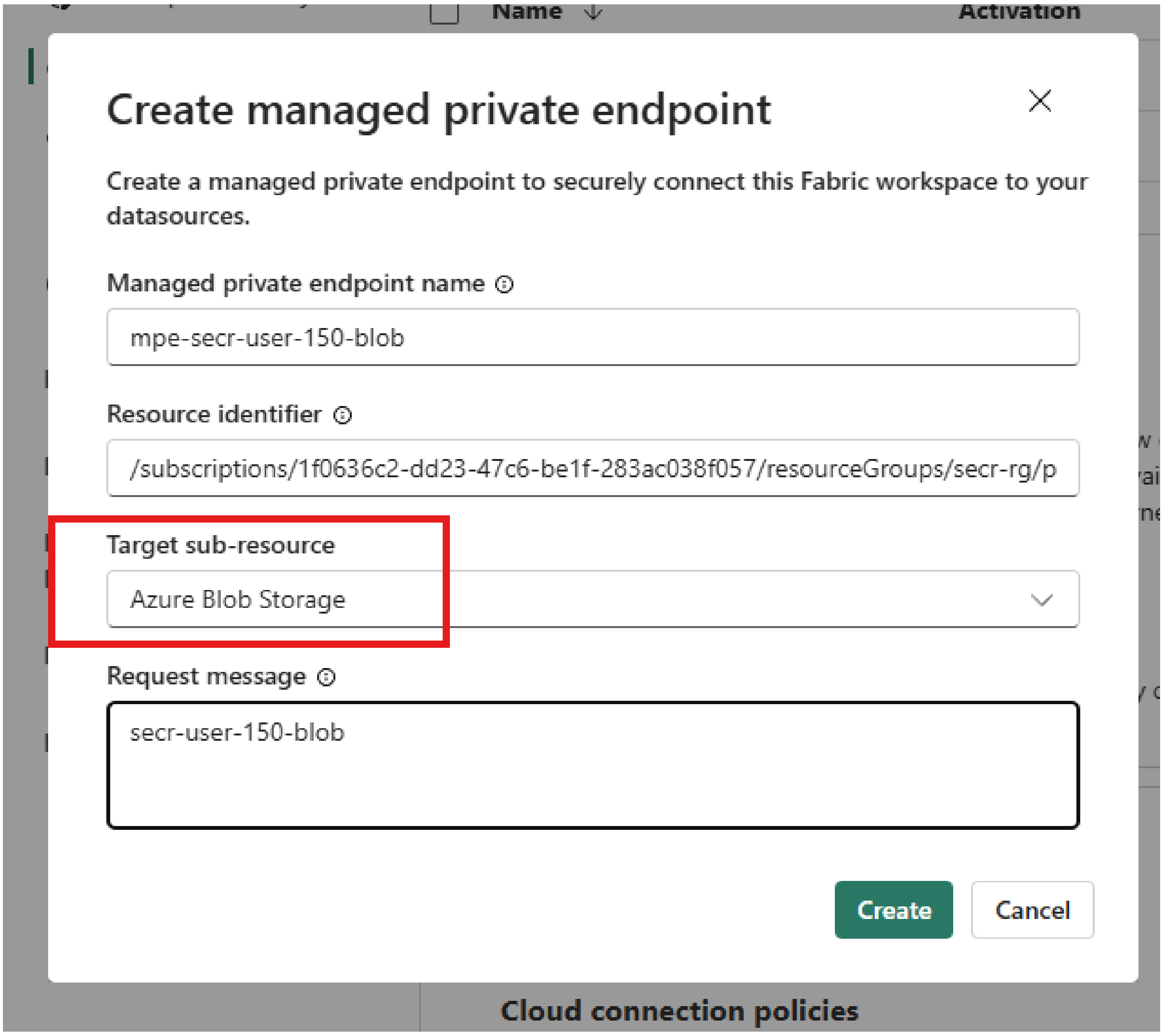
14. For **Azure Storage**, you must create separate managed private endpoints for Blob (*.blob.core.windows.net) and Data Lake storage (*.dfs.core.windows.net) service. You can see full list of endpoints associated with Azure Storage by navigating to **Azure portal** > **Azure Storage** > **Settings** > **Endpoints**.



You will observe separate endpoints OAP is a network-layer feature. For each external endpoint (which is the case here with Azure Storage), one must configure separate MPE for Blob and Data Lake storage endpoints. These are commonly used by Fabric for accessing data from Azure Storage. You will do that in the following step.

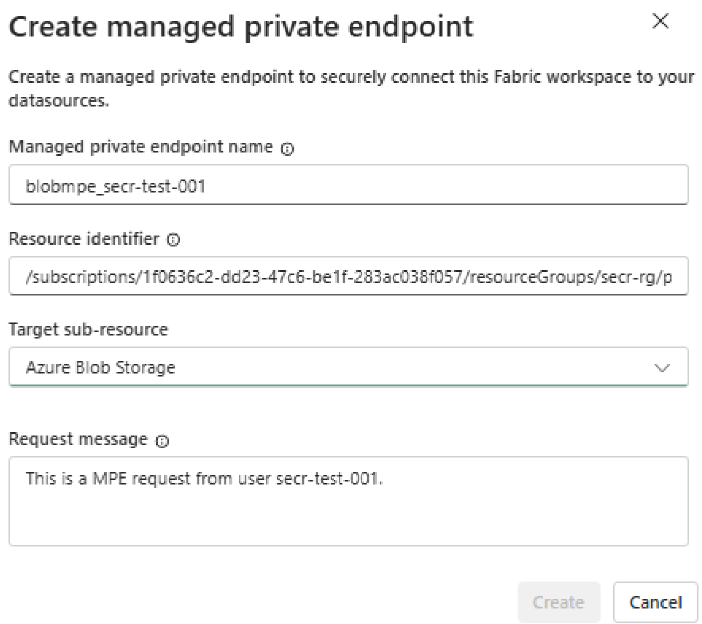
15. Create a new managed PE in addition to the one created in previous step – but this time you will choose **Azure Blob Storage for Target Sub-resource**:

    MPE Name: mpe-username-blob (Example: mpe-secr-user-150-blob)
    Resource ID: /subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/providers/Microsoft.Storage/storageAccounts/adlsfabconsec
    Target sub-resource: **Azure Blob Storage**
    Request Message: username-blob (Example: secr-user-150-blob)



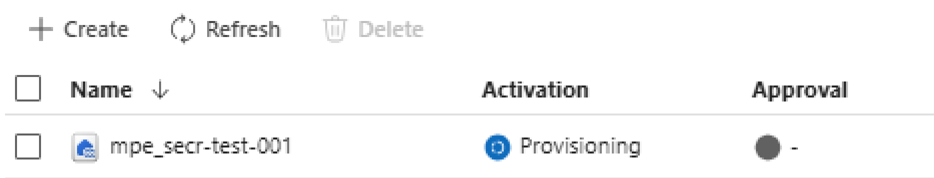
Use the following values for configuring this second MPE:

- **Managed private endpoint name:** This is a text field so you can enter any name. For this workshop, we recommend including your username so that it will be easier to identify the endpoint provisioned by you. mpe_<username>.
  Resource identifier: /subscriptions/1f0636c2-dd23-47c6-be1f-283ac038f057/resourceGroups/secr-rg/providers/Microsoft.Storage/storageAccounts/adlsfabconsec
- **Target sub-resource:** Select **Azure Blob Storage.**

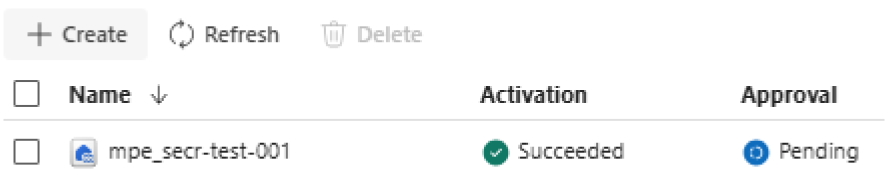


Select **Create** once all details have been entered.

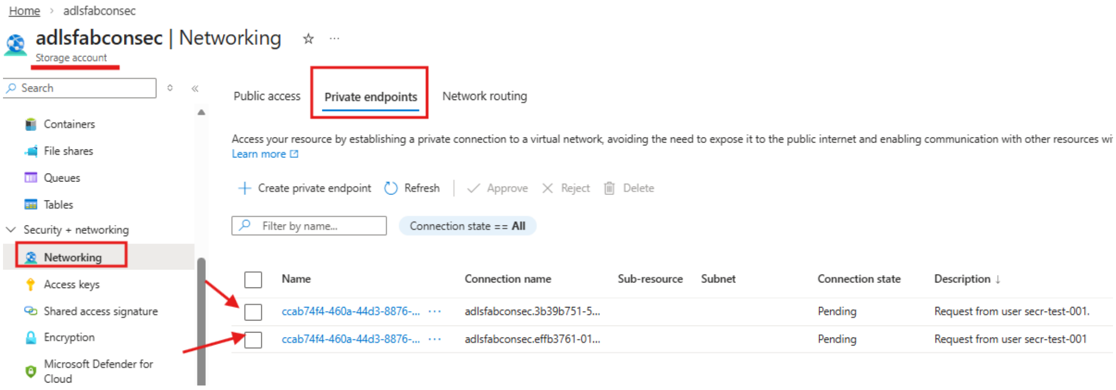
16. You should see that the **Activation** status change to **provisioning.**



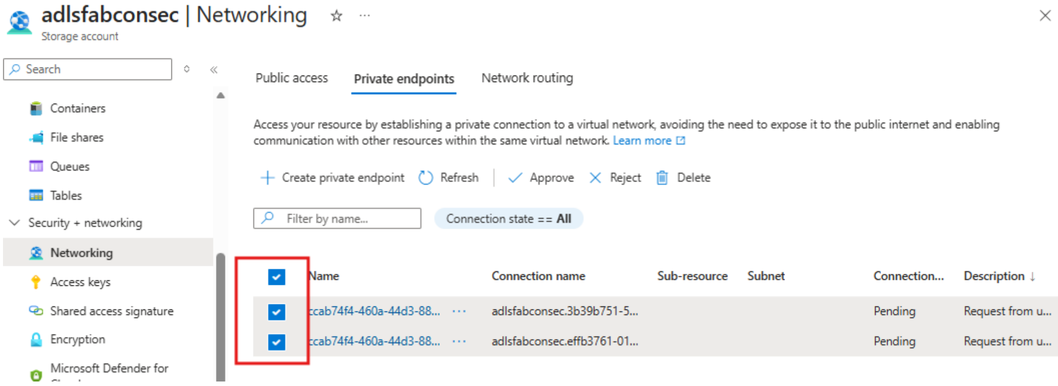
17. After 2-3 minutes, the status should change to **Succeeded.** At this stage, the resource administrator (in this case, ADLS) is ready to review and approve MPE request.



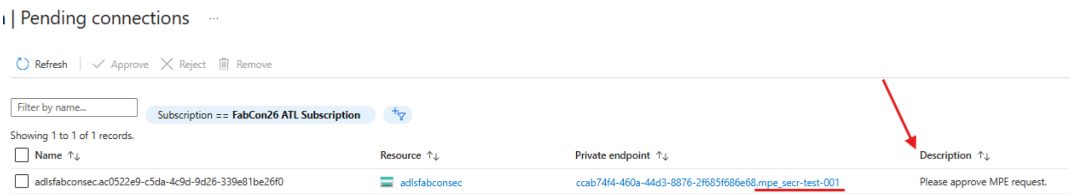
18. Switch to **Azure portal** and login using username and password provisioned for you for this workshop. Navigate to **Storage account > Networking > Private endpoints.** Here, you should see 2 requests for private endpoints in **Pending** state.



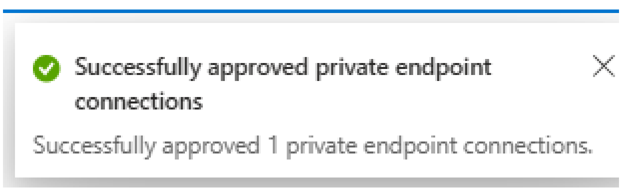
19. Select the two requests for your user/workspace and then select **Approve.**



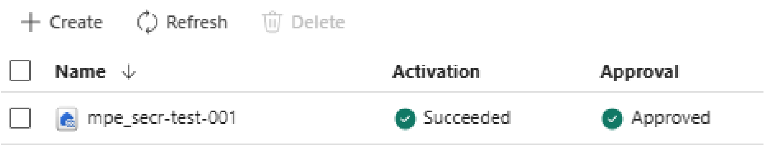
20. Since you are sharing this environment with other users, **Pending Connections** will show all the connection requests submitted by other users. Please identify the connection requested by you and approve it. In this case, we used **adlsmpe-_secr-test-001** as the name. Select the request which was submitted by you and click **Approve**.



21. Once approved, you should see a pop-up on the top-right corner inside Azure portal.

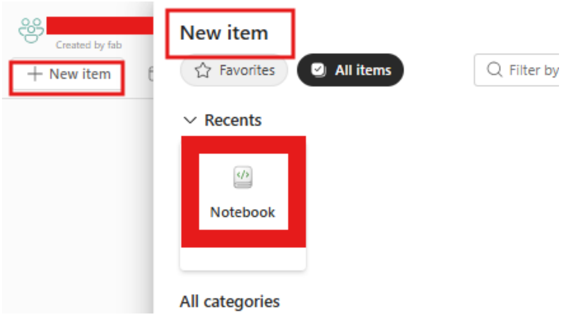


22. Switch back to Fabric portal and navigate to the workspace ending with _oap. Under **Workspace settings > Outbound networking**, you will observe that **Approval** status has changed to Approved.

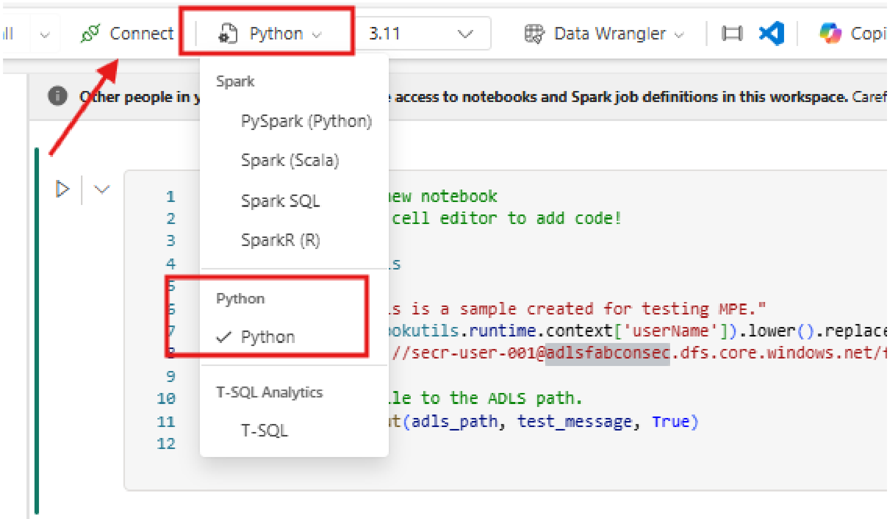


23. Using a notebook, you will now attempt to connect to ADLS from the workspace where OAP has been enabled.

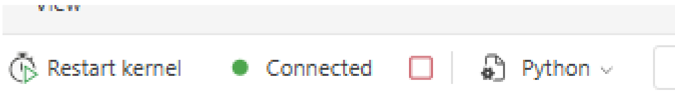
24. In Fabric workspace ending with *_oap* (where you enabled outbound access protection), create a new notebook (**or you can use the notebook/code which you created earlier at step 6**).



25. Please ensure that you are using **Python**. This has smaller footprint compared to Spark. Once you have switched to Python, select **Connect.**



It takes a few minutes for compute to provision and start for OAP enabled workspaces. You should see status switch from **Connecting …** to **Connected** when it's ready to go.



26. In the notebook cell, paste this code. Ensure that you replace variable values prior to running the code. This sample code will create a file on ADLS. If MPE for ADLS has been configured correctly, you should be able to write a file out to ADLS.

```
import notebookutils

container = "secr-user-150" #replace this with your username
storageaccount = "adlsfabconsec"
directory = "fabric_mpe_test"

test_message = "This is a sample created for testing MPE."
file_name = (notebookutils.runtime.context['userName']).lower().replace("
","") + ".txt"
adls_path =
f"abfss://{container}@{storageaccount}.dfs.core.windows.net/{directory}/{file_
name}"

#This will write file to the ADLS path.
```

```
notebookutils.fs.put(adls_path, test_message, True)
```

Replace the parameters with the following values:

- **<container>:** Container pre-created for this workshop. This should be same as your username.
- **<storageaccount>:** *adlsfabconsec*
- **<directory>**: *fabric_mpe_test*

The code should print out **True** once it has completed running. You can confirm that the file has been written by switching to **Azure Portal** and navigating to the storage account.

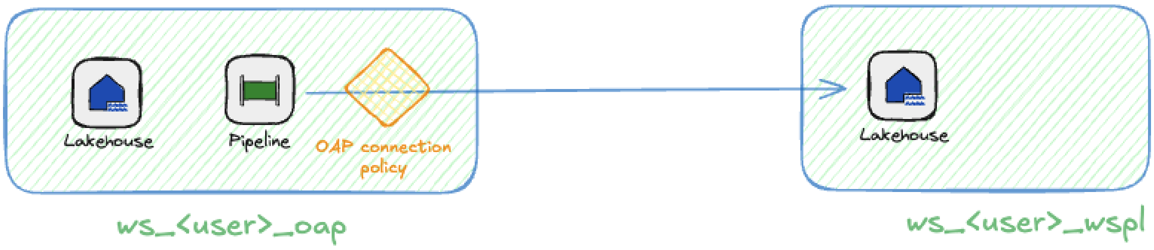
Checkpoint: At this stage, you have:

- Enabled OAP on your workspace ending with *_oap*.
- Created a managed private endpoint for ADLS and approved it via Azure Portal.
- Tested outbound connectivity to ADLS with and without MPE

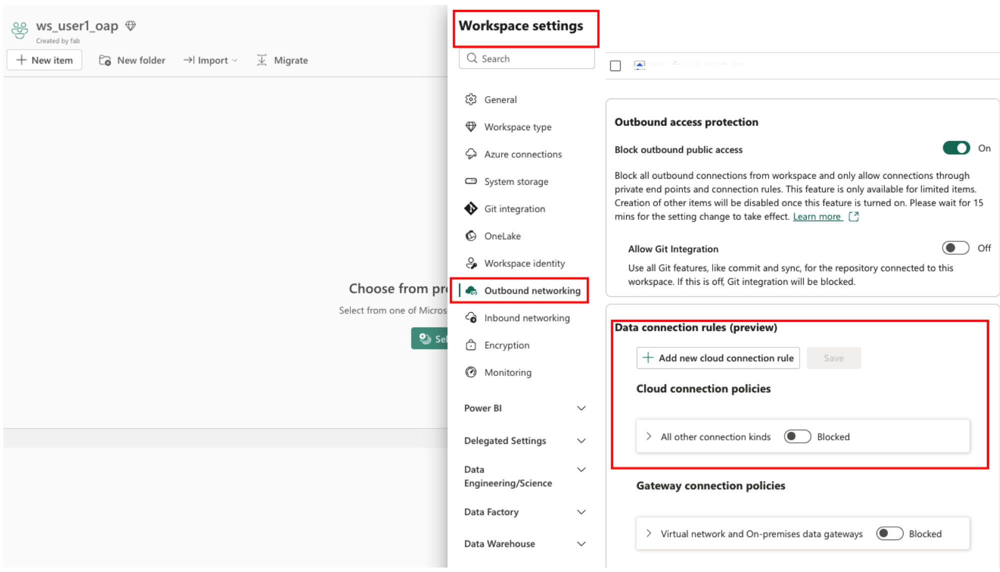
Step 2: Setup connection rules for Data Factory workloads

When outbound access protection is enabled for a workspace, all outbound connections are blocked by default. For Data Factory workloads, you can permit access to external data sources by configuring data connection rules.

In this exercise, we will be using **cloud connections** to enable outbound access for Data Factory workloads (Pipelines, Dataflows gen2). You will setup policy to allow pipeline from OAP enabled workspace to write data out to a lakehouse in **ws_<user>_public** workspace.

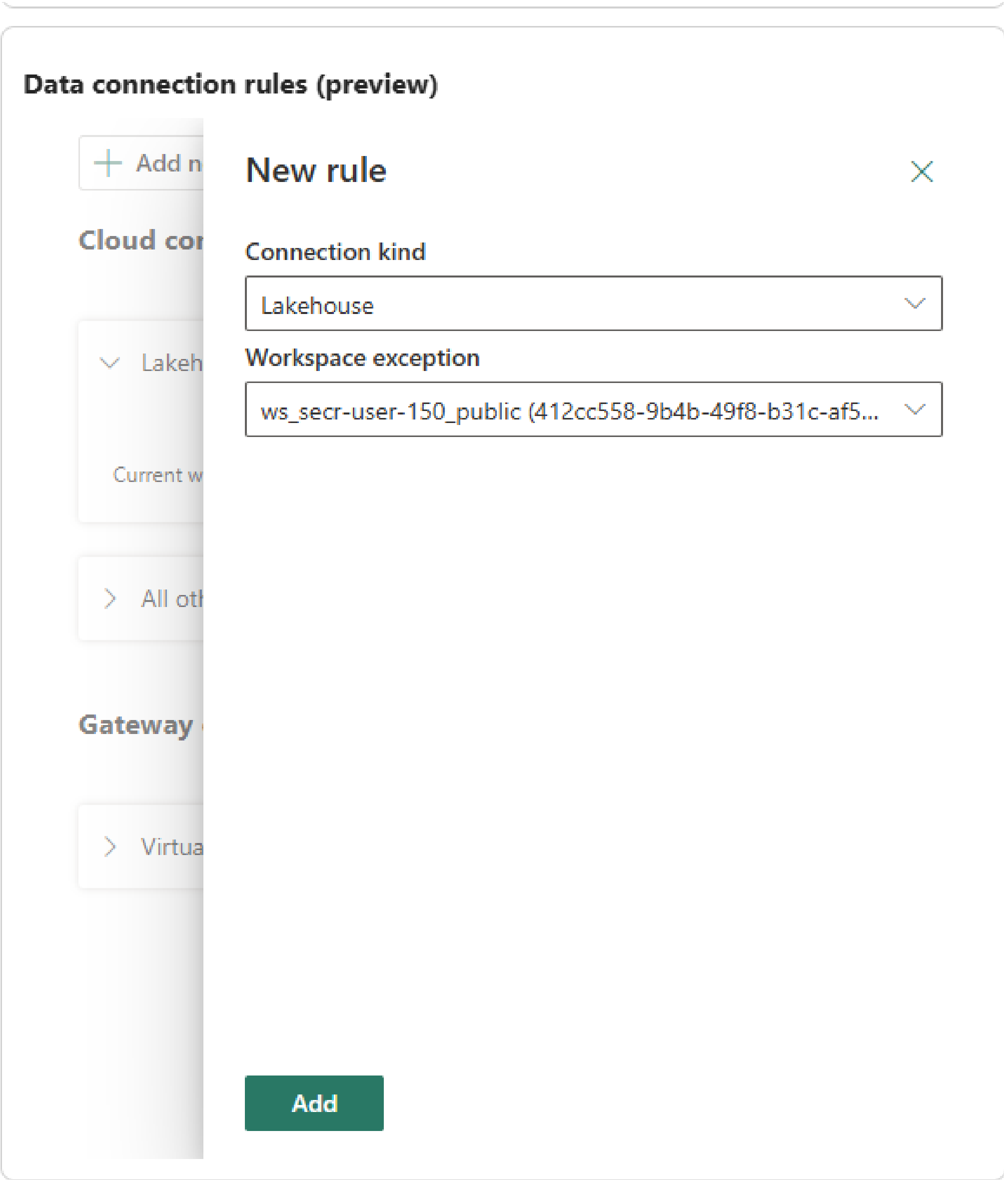


27. Navigate to workspace **ws_<user>_oap** and select **Workspace Settings > Outbound networking. B**

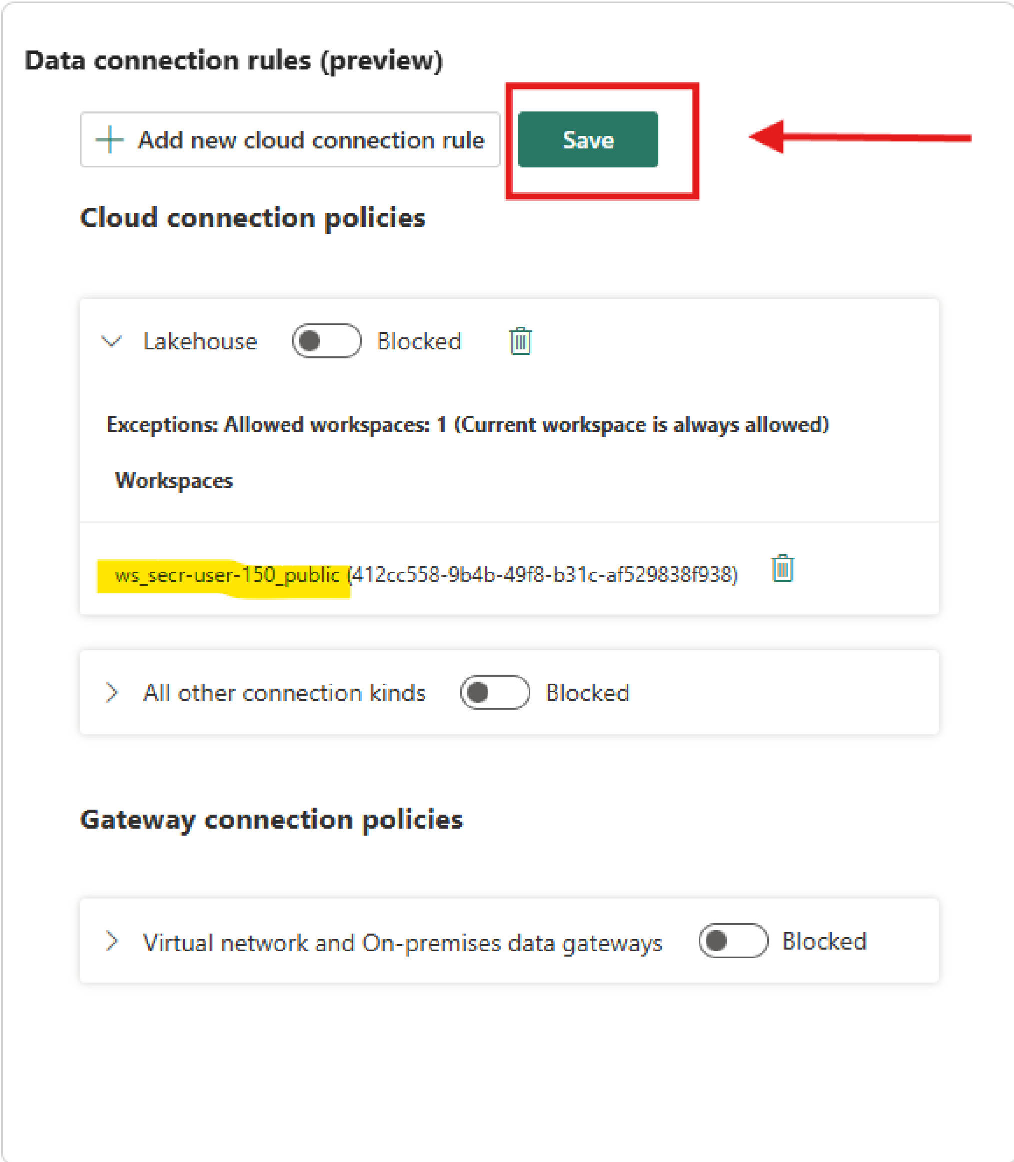


28. What you are doing here is that you are allowing outbound to lakehouse in workspace **ws_<user>_wspl**. Enter the following details.

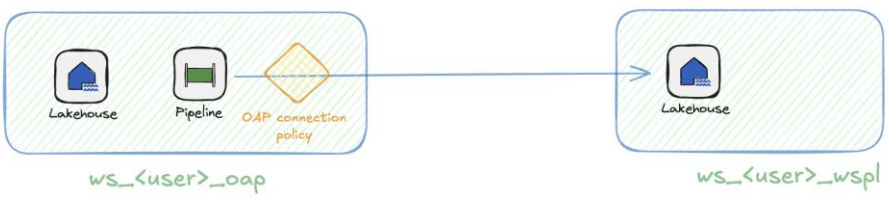
- **Connection kind**: Lakehouse
- **Workspace exception**: *Enter GUID of ws_<user>wspl workspace*



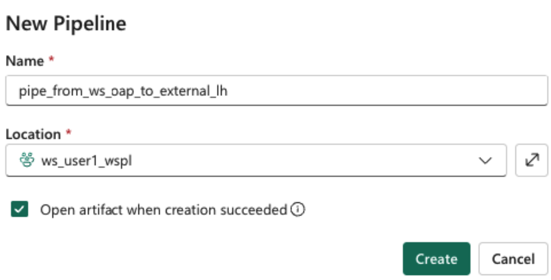
29. If configured correctly, you should see workspace ws_<user>_wspl under **Connection Policies**. Select **Save** to apply these settings.



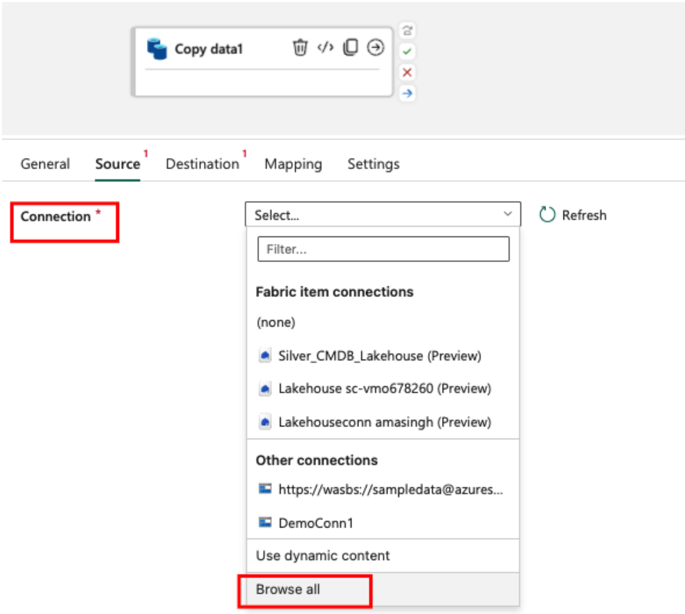
30. After the connection policy is saved, create a pipeline that **reads** from workspace **ws_<user>_oap** and writes the output to a lakehouse table in external workspace **ws_<user>_wspl.**



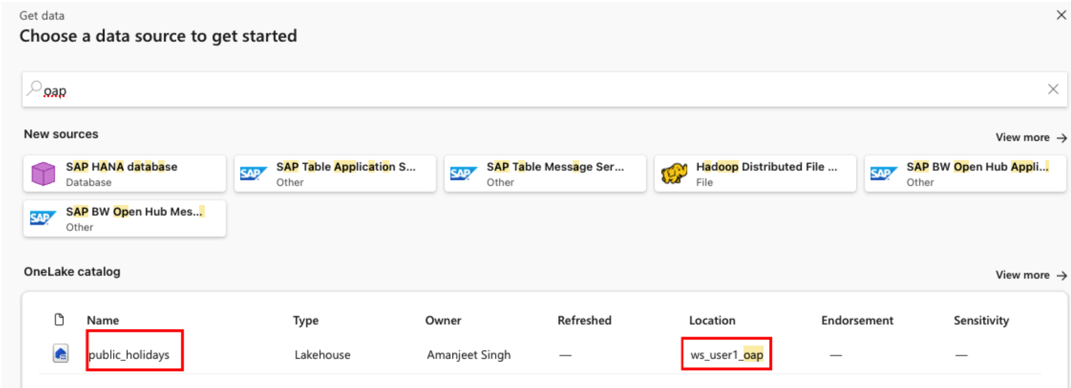
31. Create a new **Pipeline** in OAP workspace **ws_<user>_oap** assigned to you. Please replace **<user>** with the user ID assigned to you for the workshop.



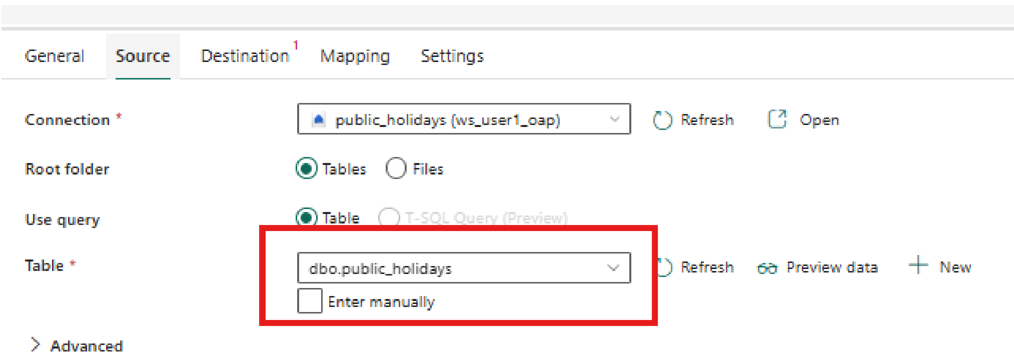
32. In Pipeline authoring experience, select the table created in the OAP workspace **ws_<user>_oap** as the source and the lakehouse **ws_<user>_public** as the target.



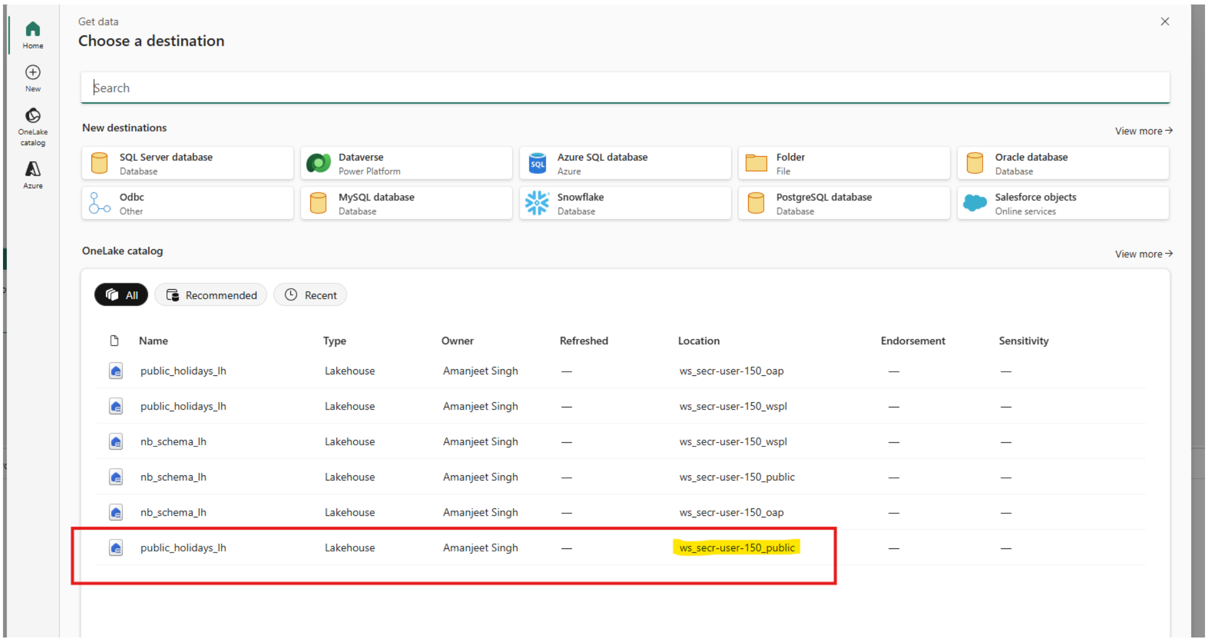
33. Select the source **lakehouse**.



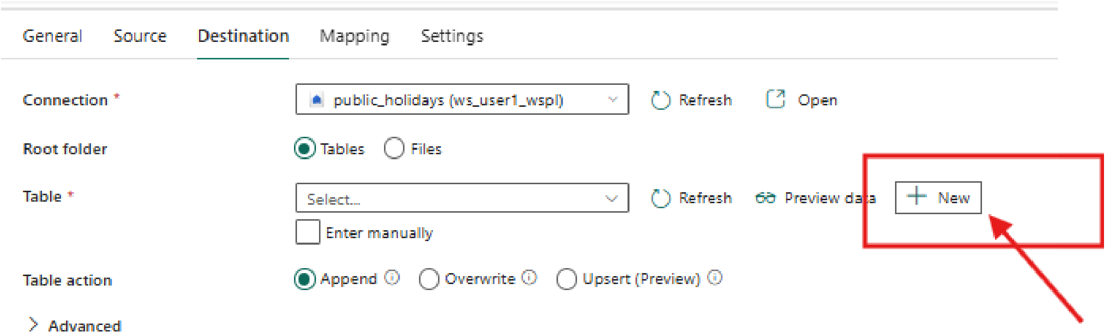
34. Select table **publicholidays.**



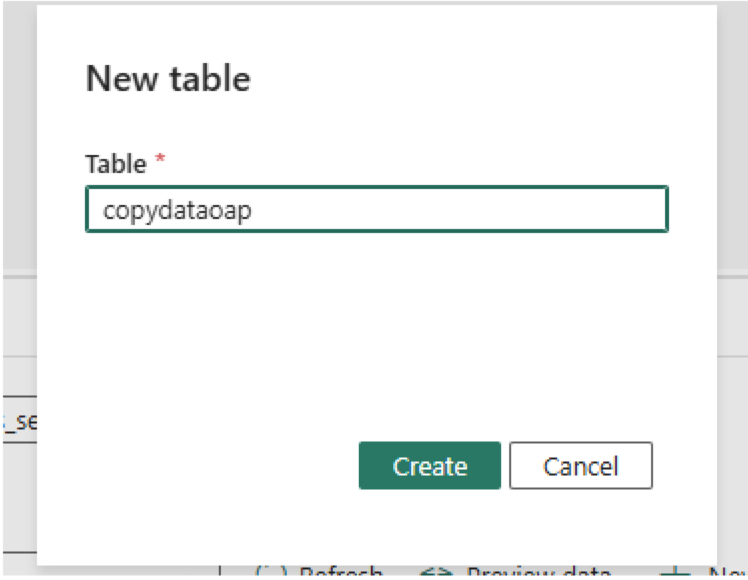
35. Select **destination** which is the pre-provisioned lakehouse in **ws_<user>_public** workspace.



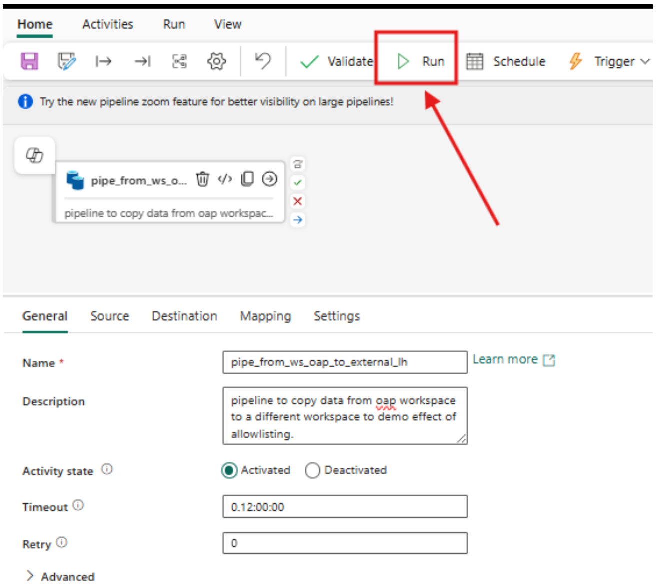
36. Select **+New** table. This table is the destination for data copied from the restricted workspace **ws_<user>_oap**, where outbound data movement is blocked. Because the target lakehouse is allowlisted via **Connection Policy**, Fabric permits writing data to this external lakehouse.



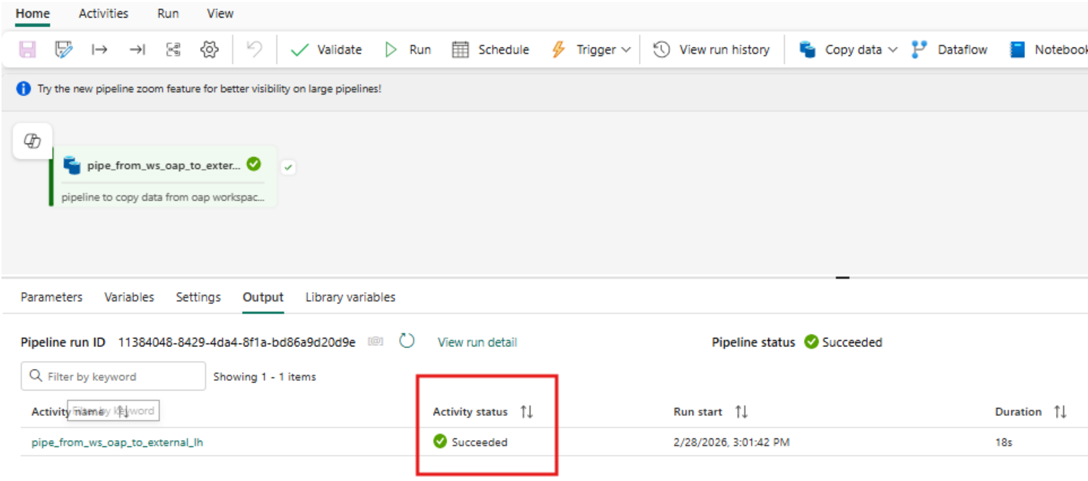
37. Enter the details of the target table and select **Create**. Please ensure that you enter appropriate details for the target table. In this example, we used schema **dbo** and table name **copydataoap**.



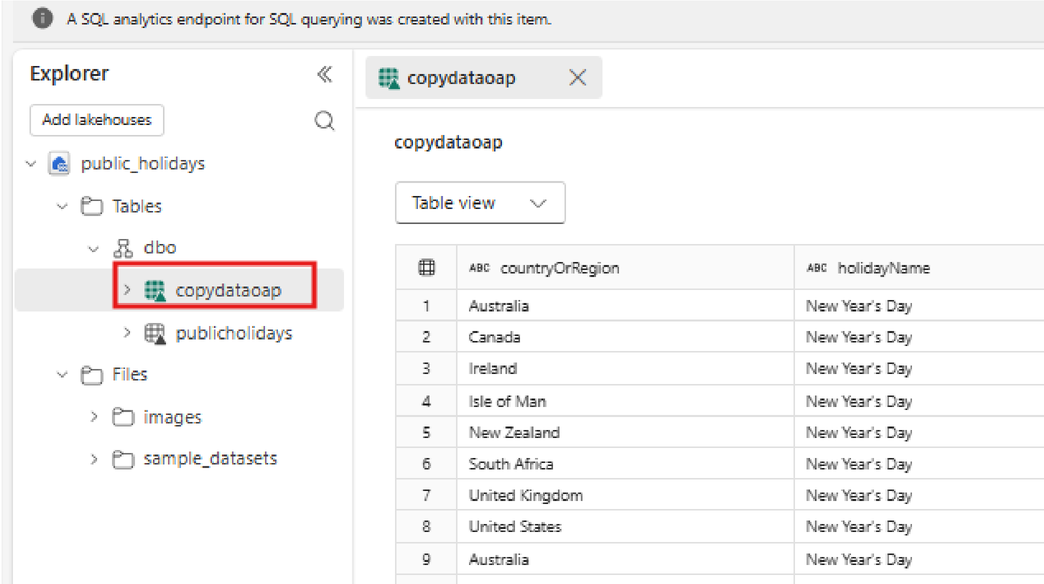
38. Select **Run** to execute the pipeline.



39. Monitor the pipeline activity in the **Output** pane at the bottom of the session. When execution completes, the **Activity Status** transitions from **Queued** → **In Progress** → **Succeeded**.


Home
Activities
Run
View
Validate
Run
Schedule
Trigger
View run history
Copy data
Dataflow
Notebook
Try the new pipeline zoom feature for better visibility on large pipelines!
pipe_from_ws_oap_to_exter...
pipeline to copy data from oap workspac...
Parameters
Variables
Settings
Output
Library variables
Pipeline run ID 11384048-8429-4da4-8f1a-bd86a9d20d9e
View run detail
Pipeline status Succeeded
Filter by keyword
Showing 1 - 1 items
Activity name
Activity status
Run start
Duration
pipe_from_ws_oap_to_external_lh
Succeeded
2/28/2026, 3:01:42 PM
18s

40. Navigate to workspace **ws_<user>_public** and select the lakehouse. You should now see the new table which you created by initiating a pipeline in OAP restricted workspace (**ws_<user>_oap**).



This completes **Connection Policy** module which is part of OAP.

You learnt how to configure an allowlist to enable outbound connectivity from a workspace with OAP enabled, and how to use a pipeline activity to copy data to an external workspace or lakehouse.

Summary

In this module, you examined how Microsoft Fabric enforces outbound access controls at the workspace level using Outbound Access Protection (OAP). You enabled OAP to block all outbound connectivity by default and configured managed private endpoints to explicitly allow secure, private access to external Azure services.

You deployed a **managed private endpoint** from a Fabric workspace to Azure Data Lake Storage Gen2, reviewed the approval workflow on the resource owner side, and validated end-to-end connectivity using a Spark notebook.

This was followed by setting up a **Connection Policy** for Data Factory workloads such as Pipelines to communicate with external services and applications including other workspaces. You created a policy to allow connectivity to a lakehouse in second workspace so that data can be copied across successfully.

Managed PE and Connection Policy features ensure that data egress occurs only through explicitly sanctioned private connections.

This concludes Module 2 – Outbound Protection.